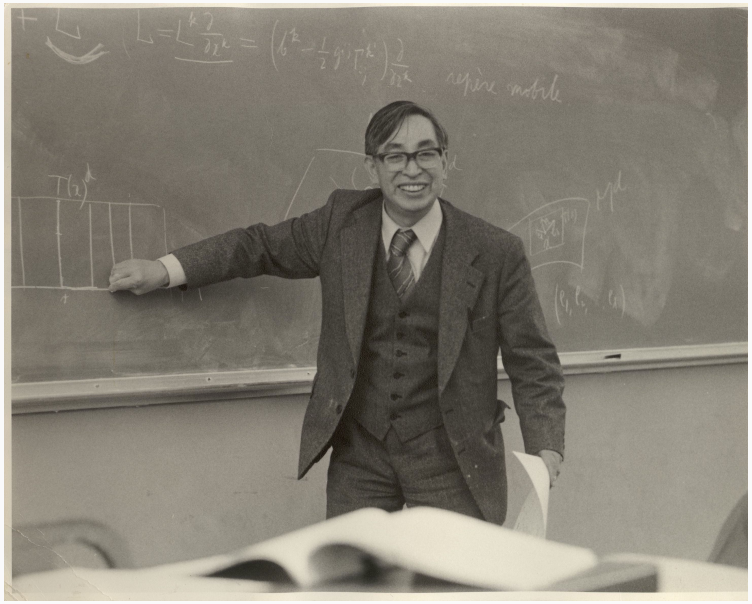
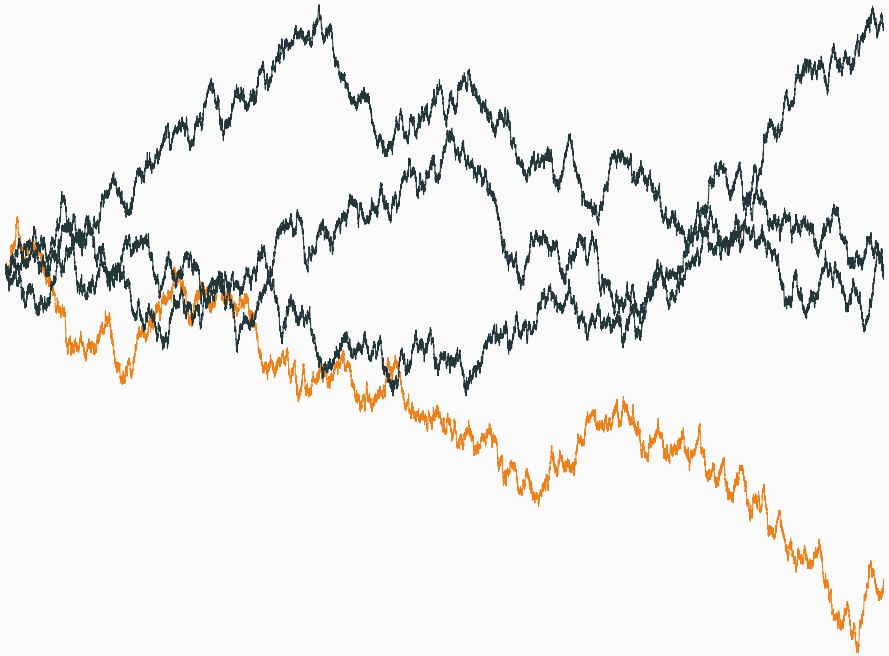
Stochastic calculus tools for quantum optics

PART I: Easy introduction to Itô calculus





Quantum Optics Seminar

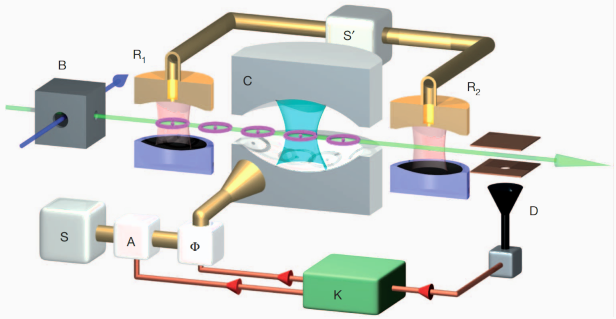
September 22, 2016

Antoine Tilloy, MPQ-theory

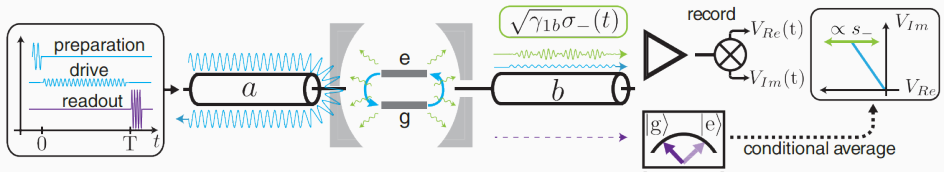
MPQ

WHY DO WE NEED STOCHASTIC CALCULUS AT ALL

It is now possible to sequentially or continuously measure the **same** quantum system and implement a feedback depending on the results.



Discrete situation: experiment of the group of Serge Haroche, Gleyzes et al. Nature **446**, 297-300 (2007)



Continuous situation: experiment of the group of Benjamin Huard, Campagne-Ibarcq et al. Phys. Rev. Lett. **112**, 180402 (2014)

Master equations $\partial_t \rho_t = \mathscr{L}(\rho_t)$ are not enough, the state itself is random $\Rightarrow \partial_t \rho_t = \mathscr{L}(\rho_t) + \text{noise}$ $(+\mathbf{feedback})$

WHY DO WE NEED TO WORK WITH STOCHASTIC CALCULUS SERIOUSLY

Playing with continuous stochastic processes as if there were regular functions gives wrong results (rigor is not the issue).

Crazy number of errors in the literature, especially in the 90's

WHAT IS ITÔ CALCULUS?

Itô calculus = extension of standard differential calculus to continuous random processes

Typically, we will give a rigorous meaning to expressions of the form:

$$\frac{\mathrm{d}f(B_t)}{\mathrm{d}t} \text{ where } \dot{X}_t = a(t) + \text{noise}$$

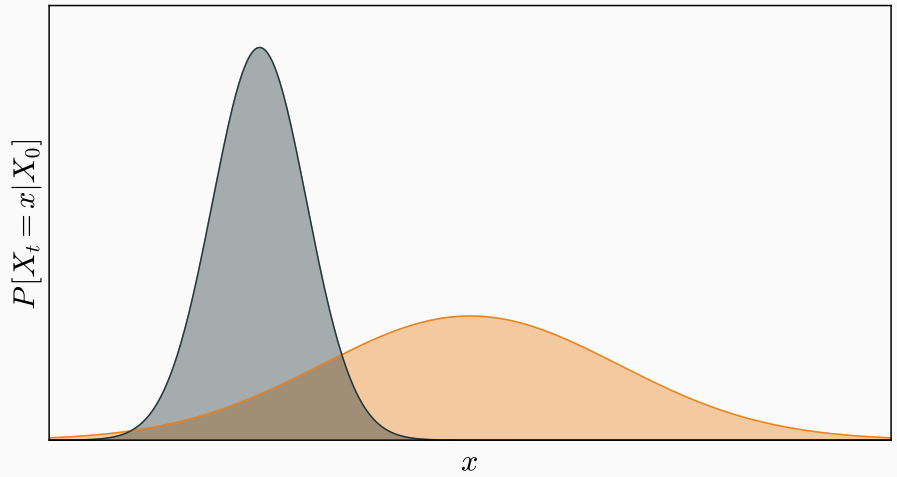
WHY USE ITÔ CALCULUS?

Physicists usually prefer other ways to deal with stochastic processes:

1. Working with deterministic equations for the probability distribution a.k.a. **Fokker-Planck** equation:

$$\frac{\partial \mathbb{P}[X_t = x|X_0]}{\partial t} = \mathscr{O} \cdot \mathbb{P}[X_t = x|X_0]$$

where typically $\mathscr{O} = -D\frac{\partial^2}{\partial x^2} + \cdots$



Rigorous **but**:

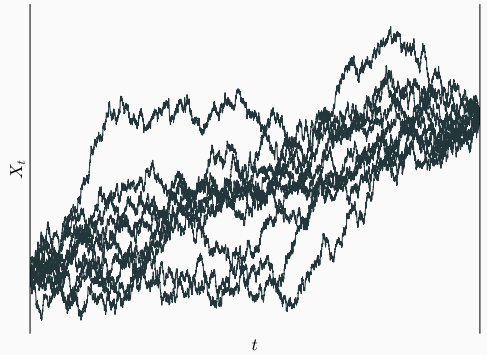
- More intuitive to work directly with the process
- $\mathscr{O}$ typically awful

WHY USE ITÔ CALCULUS?

Physicists usually prefer other ways to deal with stochastic processes:

2. Using path integrals with “Martin-Siggia-Rose”

$$\langle f(X_t)\rangle \propto \int \mathcal{D}[x]\, f(x_t)\, e^{-\int \mathrm{d}u\,(\dot{x}_u - a_u)^2}$$



Nice for perturbative expansions **but**:

- Computations sometimes awfully technical and “magical”.
- Can be ambiguous with multiplicative noise
- Fundamental properties are hidden in boundary choices of diverging determinants
- Can only be made rigorous with … Itô calculus

WHY USE ITÔ CALCULUS?

Physicists usually prefer other ways to deal with stochastic processes:

3. Using white noises carelessly a.k.a. **Langevin** equation

$$\dot{X}_t = a(t) + \sigma(t)\,\eta_t$$

with $\mathbb{E}[\eta_t] = 0$ and $\mathbb{E}[\eta_t\,\eta_s] = \delta(t-s)$.

Nice and intuitive **but**:

- Ambiguities hidden in $\delta(0)$
- Almost all standard operations one would want to do inexplicably fail.

WHY ITÔ CALCULUS

Itô calculus is a way to work rigorously with the Langevin equation

Nice aspects

- Rigorous
- Intuitive
- Emphasizes the structure of information

Problematic aspects

- Often introduced with a heavy σ-algebraic framework
- Was developed with finance in mind, looked at suspiciously by physicists

MARTINGALES AND BROWNIAN MOTION

POINT OF VIEW

Trader

We shall take the point of view of a **trader** observing the **price** X_t of a continuously quoted stock, say Alphabet.



Ewan Mc Gregor playing the role of Nick Leeson in Rogue Trader.



Useful for understanding

- the structure of information
- the concept of **martingale**

Filtration

The filtration $\{\mathcal{F}_t\}_{t\in\mathbb{R}^+}$ mathematically encodes the progressive acquisition of information. Intuitively, $\mathcal{F}_t$ is the information available to someone at time t.

Technically, "$\{\mathcal{F}_t\}$ is a filtration" $\Leftrightarrow$ "$\{\mathcal{F}_t\}$ is a set of σ-algebras such that $t \leq s \Rightarrow \mathcal{F}_t \subseteq \mathcal{F}_s$"

$\Rightarrow \mathcal{F}_t$ contains more and more events as time flows.

Examples of filtrations:

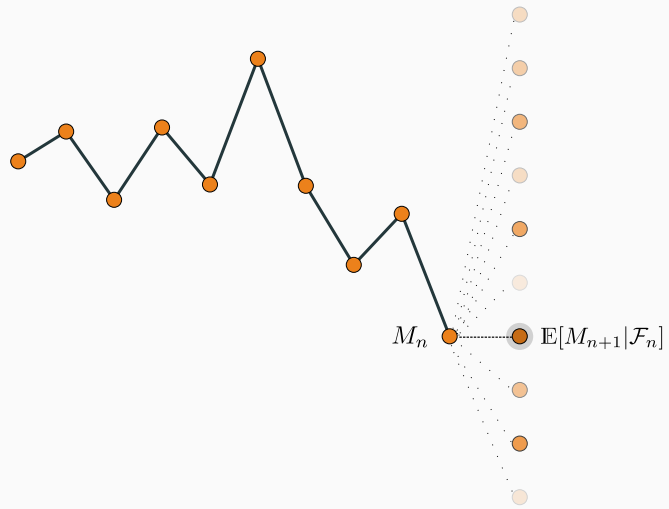
- $\mathcal{F}_t$ = "the price history of Alphabet up to time t"
- $\mathcal{G}_t$ = "all the public information in the world about Alphabet up to time t"
- $\mathcal{H}_t$ = "the price of Alphabet exactly at time t" ? $\rightarrow$ **not** a filtration

Martingale

A discrete martingale M_n is a random process such that:

$$\mathbb{E}[M_{n+1}|\mathcal{F}_n] = M_n \tag{1}$$

A martingale is a random process with increments **unforeseeable** on average.



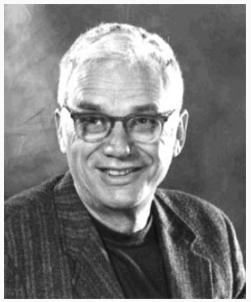
Examples of martingales:

- An unbiased random walk
- The price of Alphabet at time t with $\mathcal{F}_t =$ "stock price history up to t"
- The price of Alphabet at time t with $\mathcal{G}_t =$ "all the public information in the world before t"

Doob decomposition

Any stochastic process X_n [adapted to a filtration $\mathcal{F}_n$] can be written as the sum of a predictable process D_n and a martingale M_n:

$$X_n = \underbrace{D_n}_{\text{predictable}} + \underbrace{M_n}_{\text{martingale}}$$



Joseph L. Doob

Martingales are the universal “core” of stochastic processes

Stopped martingale

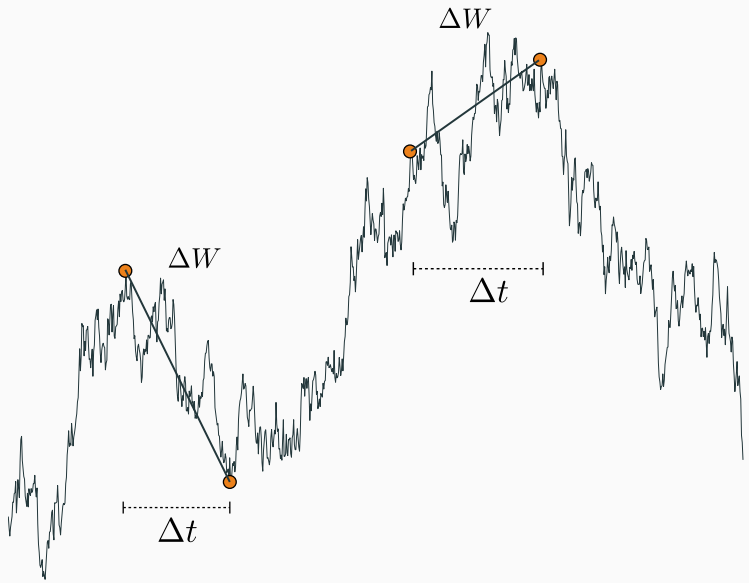
A martingale conditioned to stop when some event happens [in a time finite on average]is still a martingale.

Encodes the fact that you cannot make money by trading a martingale.

BROWNIAN MOTION

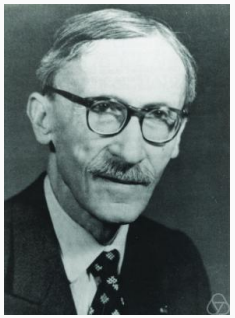
Formal definition

The Wiener process W_t is **the** continuous stochastic process with independent Gaussian increments ΔW of law $\mathcal{N}(0, \sqrt{\Delta t})$.

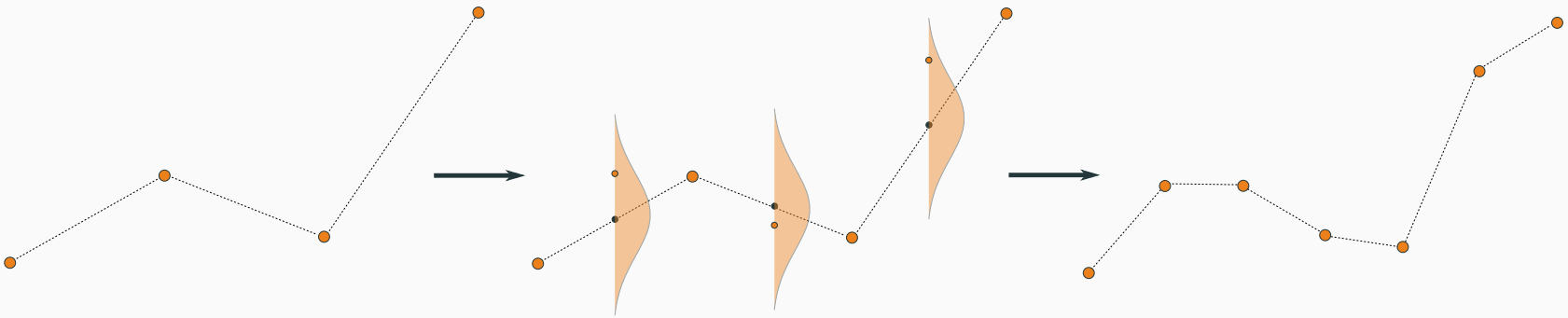


Lévy's construction

Take a discrete process X_i with Gaussian increments of law $\mathcal{N}(0, \sigma)$. Refine it by drawing new random points $X_{i+\frac{1}{2}}$ of law $\mathcal{N}\left(\frac{X_{i+1}+X_i}{2}, \frac{\sigma}{\sqrt{2}}\right)$. Iterate



Paul Lévy



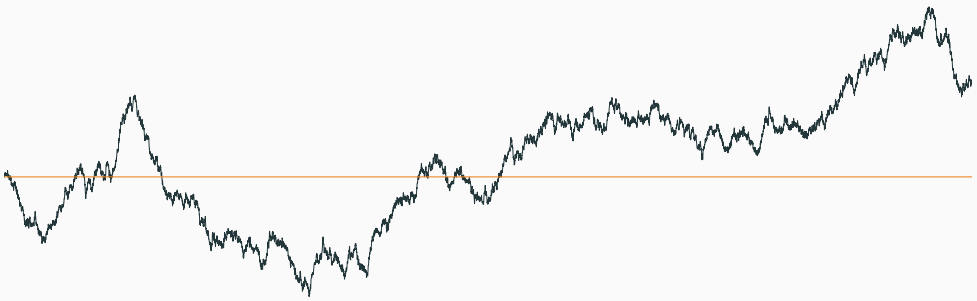
Useful for numerics to dynamically refine a mesh in a simulation involving a Brownian motion.

PROPERTIES

1. It is a martingale $\rightarrow$ trivial
2. Hausdorff dimension $\frac{3}{2}$



3. Return time in zero



WHY IS THE BROWNIAN MOTION SO STANDARD

Martingale representation theorem

Continuous martingales are Brownian motions up to a time reparametrization.

Summary "Martingales & Brownian motion"

- Martingales are random processes that are unforeseeable on average.
- Martingales have nice properties and by studying them we study all interesting processes (Doob decomposition)
- The Brownian motion is a particular kind of continuous martingale
- Actually all continuous martingales are related to the Brownian motion

Stochastic integral and Itô's lemma

Basic problem

The Brownian motion cannot be differentiated in the standard way

$$\Delta W \sim \sqrt{\Delta t} \Rightarrow \frac{\Delta W}{\Delta t} \propto \frac{1}{\sqrt{\Delta t}}$$

The **white noise** $\eta \stackrel{?}{=} \frac{\mathrm{d}W}{\mathrm{d}t}$ is ill defined.

- No derivative either in the weak sense in Sobolev spaces
- There exists elaborate ways to define it with Bochner-Milnos theorem $\rightarrow$ useful only in $D \geq 2$ where Itô doesn't work.

There is a way! One can define $\forall f$:

$$\mathscr{I}_f = \underbrace{\int f(t)\,\mathrm{d}\,W_t}_{\text{Itô integral}} \quad \text{“}=\text{”} \quad \int f(t)\,\frac{\mathrm{d}\,W_t}{\mathrm{d}t}\,\mathrm{d}t$$

We will then omit the integral sign for convenience and have:

$$f(t)\,\mathrm{d}\,W_t$$

Definition

For a stochastic process f with mild assumptions we define:

$$\int_0^T f(t)\,\mathrm{d}\,W_t := \lim_{N\to+\infty} \sum_{k=0}^{N} f\left(\frac{T}{N}\,k\right)\left[W\left(\frac{T}{N}(k+1)\right) - W\left(\frac{T}{N}\,k\right)\right]$$

- The indices are important: the Brownian increment is infinitesimally in the **future** of f.

It works essentially because the Brownian motion has **finite quadratic variation**:

$$\sum_{k=0}^{N} \left[W\left(\frac{T}{N}(k+1)\right) - W\left(\frac{T}{N}k\right) \right]^2 \underset{N\to+\infty}{\longrightarrow} [W]_T = T$$

Take the square, use Cauchy-Schwartz inequality, the previous discretized Itô integral is bounded, in the limit, in the following way:

$$\left| \int f(t) \, \mathrm{d}\, W_t \right| \leq \sqrt{[W]_T} \, \|f\|_\infty$$

This **does not** proves convergence but at least shows it doesn't blow up.

We can now write stochastic differential equations:

$$\mathrm{d}X_t = \underbrace{\mu(X_t, t)\,\mathrm{d}t}_{\text{predictable “drift”}} + \underbrace{\sigma(X_t, t)\,\mathrm{d}\,W_t}_{\text{noise}},$$

which is a **notation** for:

$$X_t = \underbrace{\int^t \mu(X_u, u)\,\mathrm{d}u}_{\text{standard integral}} + \underbrace{\int^t \sigma(X_u, u)\,\mathrm{d}\,W_u}_{\text{Itô integral}}$$

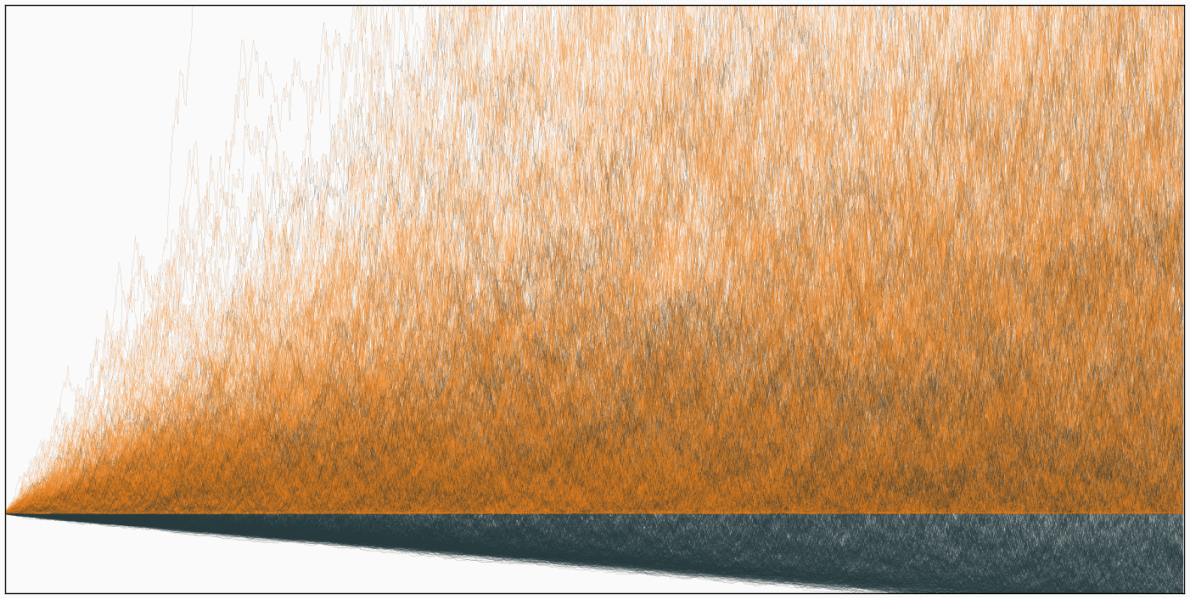
X_t is called an **Itô process**. Note that it provides an explicit Doob decomposition of X_t.

ITÔ'S LEMMA

The standard rules of differentiations don't work with this new integral:

$$\mathrm{d}\, W_t^2 \overset{?}{=} 2\, W_t\, \mathrm{d}\, W_t$$

Impossible, W_t^2 is clearly **not** a martingale, whereas $2 \int W_t\, \mathrm{d}\, W_t$ is.



Plot of 1000 trajectories of what should be the same stochastic process: W_t^2 and $2 \int W_t\, \mathrm{d}\, W_t$

ITÔ'S LEMMA

Lemma

Let $f \in \mathcal{C}^2$ and X_t an Itô process s.t.:

$$\mathrm{d}X_t = \mu(X_t, t)\,\mathrm{d}t + \sigma(X_t, t)\,\mathrm{d}W_t,$$

then:

$$\mathrm{d}f(X_t, t) = \underbrace{\frac{\partial f}{\partial t}\,\mathrm{d}t + \frac{\partial f}{\partial X}\,\mathrm{d}X_t}_{\text{trivial}} + \underbrace{\frac{1}{2}\frac{\partial^2 f}{\partial X^2}\,\sigma(X_t, t)^2\,\mathrm{d}t}_{\text{Itô correction}}$$

- $f \in \mathcal{C}^2$ can be relaxed $\rightarrow$ Tanaka's formula

Proof: Take the definition of the Itô integral and Taylor expand everything until the remainder is of order $\mathcal{O}(\mathrm{d}t^{3/2})$.

Itô's lemma

Physicist's cookbook

One can work with Itô processes by using only a few rules:

- $|\mathrm{d}\, W_t| \sim \sqrt{\mathrm{d}t}$
- $\mathbb{E}[\mathrm{d}\, W_t | \mathcal{F}_t] = 0$
- $\mathrm{d}\, W_t \, \mathrm{d}\, W_t = \mathrm{d}t \ \simeq$ Itô's lemma

SUMMARY “ITÔ INTEGRAL”

- We can’t define the derivative of the Brownian but can define integrals with respect to the Brownian motion
- The rules of differentiation slightly change
- In the end, one only needs a minimal knowledge to apply the formalism

ADDITIONAL SUBTLETIES

ADDITIONAL SUBTLETIES

1. Stratonovitch integral
2. Wong-Zakaï theorem

Definition

For a stochastic process f with mild assumptions we define:

$$\int_0^T f(t) \circ \mathrm{d}W_t := \lim_{N \to +\infty} \sum_{k=0}^{N} \frac{f\left(\frac{T}{N} k\right) + f\left(\frac{T}{N}(k+1)\right)}{2} \left[W\left(\frac{T}{N}(k+1)\right) - W\left(\frac{T}{N} k\right) \right]$$

- The indices are important: the Brownian increment is now **correlated** with f.

STRATONOVITCH INTEGRAL

Itô and Strato are related:

$$f(W_t, t) \circ \mathrm{d}W_t = f(W_t, t)\,\mathrm{d}W_t + \frac{1}{2}\frac{\partial f}{\partial W}(W_t, t)\,\mathrm{d}t$$

Strato is convenient because:

- The standard rules of differentiation apply
- It is robust to smoothing

Itô is convenient because:

- $\mathrm{d}W_t$ is decorrelated from the past
- Averaging evolutions is trivial

Wong-Zakaï theorem

Let W_t^ε be a smoothed version of W_t, then:

$$\underbrace{f(t)\,\mathrm{d}\,W_t^\varepsilon}_{\text{Standard integral}} \underset{\varepsilon\to 0}{\longrightarrow} \underbrace{f(t)\circ\mathrm{d}\,W_t}_{\text{Strato integral}}$$

Most of the mistakes in the literature come from this!